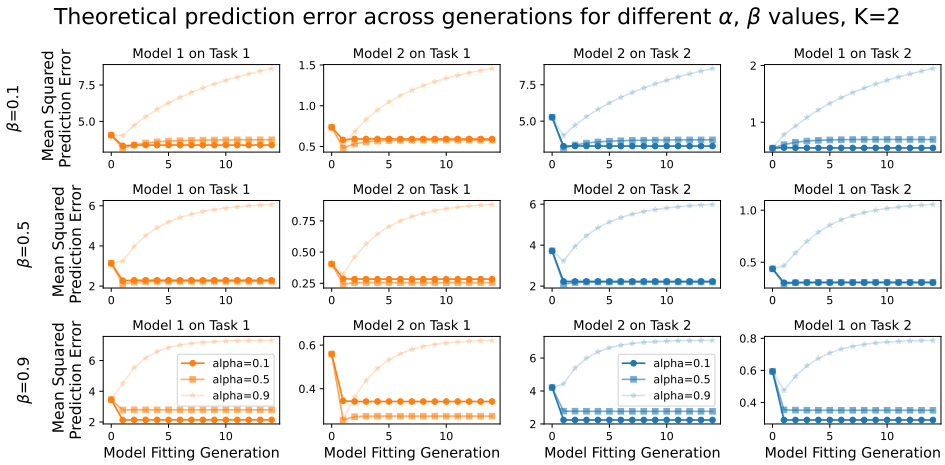

Figure 4: **Predicted behavior over time for a $K = 2$ model system with varying α, β.** *We use equations for MSE from theorem 2 and run simulations with $K = 2$, dimension 50, rank 15.*

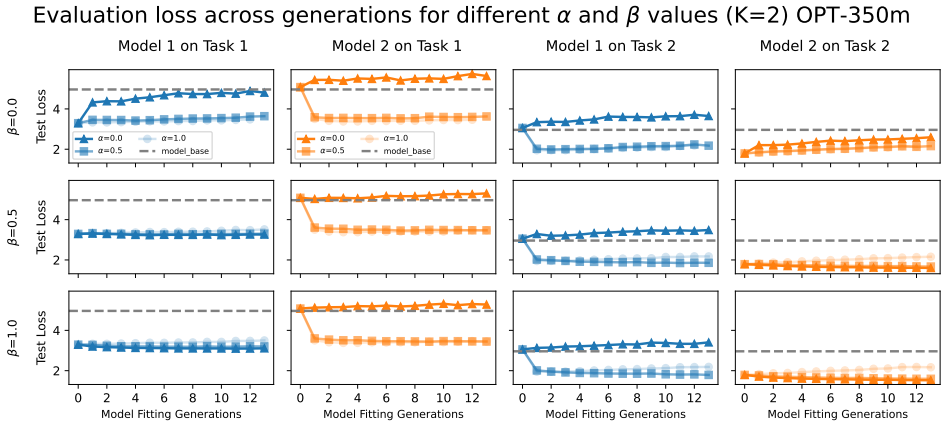

Figure 5: **Actual behavior over time for interactions between OPT models ($K = 2$) with varying α, β.**

and for the public data (X_*, y_*). At each generation t, the data (X_{tk}, y_{tk}) produced by entity k is given by

$$y_{tk} = X_{tk}\hat{\theta}_{t-1,k} + w_{tk}$$

where $\hat{\theta}_{t-1,k} \in \mathbb{R}^d$ is the most recent parameter estimate and $w_{tk} \sim \mathsf{N}(0, \sigma^2 \mathrm{I})$ is Gaussian noise that is independent across entities and generations. To represent the notation compactly, we use the embedding

$$\tilde{\boldsymbol{X}} = \begin{bmatrix} \tilde{X}_1 & & \\ & \ddots & \\ & & \tilde{X}_K \end{bmatrix}, \quad \boldsymbol{y}_0 = \begin{bmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_K \end{bmatrix} \quad \boldsymbol{X}_t = \begin{bmatrix} X_{t1} & & \\ & \ddots & \\ & & X_{tK} \end{bmatrix}, \quad \boldsymbol{y}_t = \begin{bmatrix} y_{tk} \\ \vdots \\ y_{tK} \end{bmatrix}, \quad \boldsymbol{w}_t = \begin{bmatrix} w_{t1} \\ \vdots \\ w_{tK} \end{bmatrix},$$

Linear dynamical system According to the workflow, each parameter estimate is obtained as the minimizer in $\theta \in \mathbb{R}^d$ of the empirical loss:

$$\bar{\alpha}_t\beta_t\|\tilde{y}_k - \tilde{X}_k\theta\|^2 + \bar{\alpha}_t\bar{\beta}_t\|y_* - X_*\theta\|^2 + \frac{\alpha_t}{K}\sum_{j=1}^{K}\|y_{tj} - X_{tj}\theta\|^2,$$

where $\alpha_0 \equiv 0$ and so only the first two terms are present at initialization. The minimum norm solution is given in closed form by

$$\hat{\theta}_{tk} = \left(\bar{\alpha}_t\beta_t\tilde{S}_k + \bar{\alpha}_t\bar{\beta}_t S_*^\top + \frac{\alpha_t}{K}\sum_{j=1}^{k} S_{tk}\right)^+ \left(\bar{\alpha}_t\beta_t\tilde{X}_k^\top\tilde{y}_t + \bar{\alpha}_t\bar{\beta}_t X_*^\top y_* + \frac{\alpha_t}{K}\sum_{j=1}^{k} X_{tj}^\top y_{tj}\right)$$

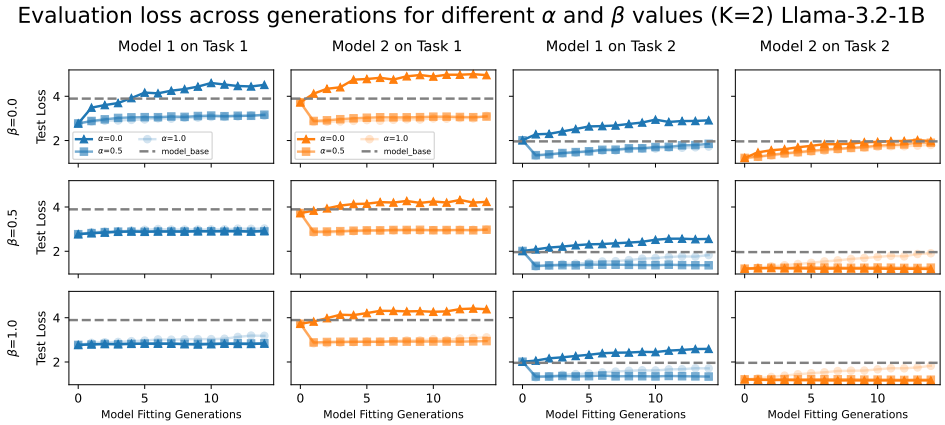

Figure 6: **Actual behavior over time for interactions between Llama models** ($K = 2$) **with varying** α, β.

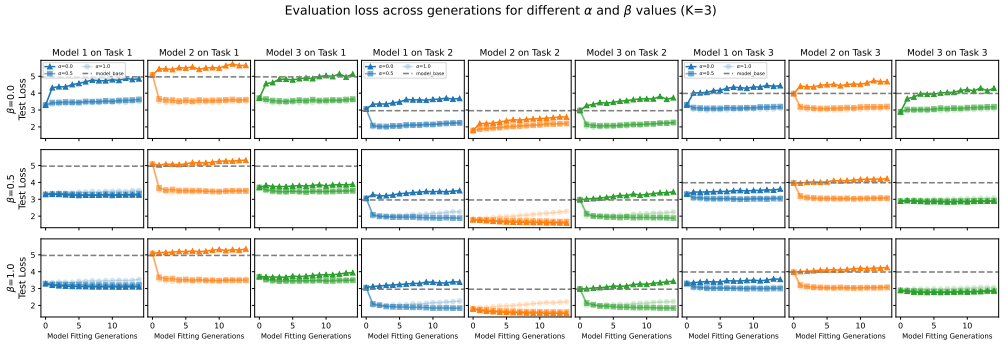

Figure 7: **Actual behavior over time for interactions between OPT models** ($K = 3$) **with varying** α, β.

where $\tilde{S}_k = \tilde{X}_t^\top \tilde{X}_k$, $S_* = X_* X^\top$, $S_{tk} = X_{tk}^\top X_{tk}$, and $(\cdot)^+$ denotes the Moore-Penrose pseudoinverse. Stacking the estimates into a vector $\hat{\boldsymbol{\theta}} = \mathsf{vec}(\hat{\theta}_1, \ldots, \hat{\theta}_K)$ and using the identity $A^\top = A^\top A A^+$, we can express all estimate updates simultaneously as

$$\hat{\boldsymbol{\theta}}_t = \boldsymbol{G}_t^+ \Big(\bar{\alpha}_t \beta_t \tilde{\boldsymbol{S}} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} + \bar{\alpha}_t \bar{\beta}_t (\mathbf{1}_K \otimes S_*) X_*^+ y_* + \frac{\alpha_t}{K} (\mathbf{1}_K \otimes \sum_{j=1}^{k} S_{tj} X_{tj}^+ y_{tj}) \Big), \tag{3}$$

where $\mathbf{1}_K$ denotes the $K \times 1$ vector of ones and $\boldsymbol{G}_t$ is the block diagonal matrix given by

$$\boldsymbol{G}_t \coloneqq \bar{\alpha}_t \beta_t \tilde{\boldsymbol{S}} + \bar{\alpha}_t \bar{\beta}_t (\mathrm{I}_K \otimes S_*) + \alpha_t (\mathrm{I}_K \otimes \underline{\boldsymbol{S}}_t), \qquad \underline{\boldsymbol{S}}_t = \frac{1}{K} \sum_{k=1}^{K} S_{kt}.$$

Defining the orthogonal projection matrix $\Pi = \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top \otimes \mathrm{I}_d$ we can write

$$\frac{1}{K} (\mathbf{1}_K \otimes \sum_{j=1}^{k} S_{tj} X_{tj}^+ y_{tj}) = \Pi \boldsymbol{S}_t \boldsymbol{X}_t^+ \boldsymbol{y}_t.$$

Introducing the matrices $\boldsymbol{P}_t \coloneqq \bar{\alpha}_t \boldsymbol{G}_t^+ \begin{bmatrix} \beta_t \tilde{\boldsymbol{S}} & \bar{\beta}_t (\mathbf{1}_K \otimes S_*) \end{bmatrix}$ and $\boldsymbol{Q}_t \coloneqq \alpha_t \boldsymbol{G}_t^+ \Pi \boldsymbol{S}_t$, can express (3) as

$$\hat{\boldsymbol{\theta}}_t = \boldsymbol{P}_t \begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix} + \boldsymbol{Q}_t \boldsymbol{X}_t^+ \boldsymbol{y}_t.$$

Using $\boldsymbol{y}_t = \boldsymbol{X}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{w}_t$ and $\boldsymbol{Q}_t \boldsymbol{X}_t^+ \boldsymbol{X}_t = \boldsymbol{Q}_t$, we obtain

$$\hat{\boldsymbol{\theta}}_t = \boldsymbol{P}_t \begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix} + \boldsymbol{Q}_t \hat{\boldsymbol{\theta}}_{t-1} + \boldsymbol{Q}_t \boldsymbol{X}_t^+ \boldsymbol{w}_t. \tag{4}$$

This expression shows that the estimates evolve according to a discrete-time linear dynamical system (also known as a Kalman filter model) with state variable $\hat{\boldsymbol{\theta}}_t$.

E.1 Proof of Theorem 1

We now derive the distribution of $\hat{\boldsymbol{\theta}}_t$ conditional on the initial data D_0 given by $(\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{y}})$ and (X_*, y_*). Specifically, we show that the estimates are Gaussian with mean and variance

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_t \mid D_0] = \boldsymbol{M}_t \begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix}, \qquad \mathsf{Cov}(\hat{\boldsymbol{\theta}}_t \mid D_0) = \boldsymbol{C}_t$$

where the matrices $\boldsymbol{M}_t$ and $\boldsymbol{C}_t$ are defined recursively with $\boldsymbol{M}_0 = \boldsymbol{P}_0$ and $\boldsymbol{C}_0 = \boldsymbol{0}_{Kd \times Kd}$ and

$$\boldsymbol{M}_t = \boldsymbol{P}_t + \boldsymbol{Q}_t \boldsymbol{M}_{t-1}, \qquad \boldsymbol{C}_t = \boldsymbol{Q}_t(\sigma^2 \boldsymbol{S}_t^+ + \boldsymbol{C}_{t-1})\boldsymbol{Q}_t, \qquad t \geq 1.$$

The proof is by mathematical induction. For the base case $t = 0$ we invoke (4) along with $\boldsymbol{Q}_0 = 0$ to see that $\hat{\boldsymbol{\theta}}_0$ is a deterministic function of the initial data with $\boldsymbol{M}_0 = \boldsymbol{P}_0$ and $\boldsymbol{C}_0 = \boldsymbol{0}_{Kd \times Kd}$.

For the inductive case, assume that the stated distribution holds up to generation $t-1$. From the definition of the workflow, (4) holds with $\boldsymbol{w}_t \sim \mathsf{N}(0, \sigma^2 \mathrm{I})$ independent of everything else. Thus $\hat{\boldsymbol{\theta}}_t$ is Gaussian with mean

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_t \mid D_0] = \boldsymbol{P}_t \begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix} + \boldsymbol{Q}_t \mathbb{E}[\hat{\boldsymbol{\theta}}_{t-1} \mid D_0] = \underbrace{(\boldsymbol{P}_t + \boldsymbol{Q}_t \boldsymbol{M}_{t-1})}_{\boldsymbol{M}_t} \begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix}$$

and covariance

$$\mathsf{Cov}(\hat{\boldsymbol{\theta}}_t \mid D_0) = \mathsf{Cov}(\boldsymbol{Q}_t \hat{\boldsymbol{\theta}}_{t-1} \mid D_0) + \mathsf{Cov}(\boldsymbol{Q}_t \boldsymbol{X}_t^+ \boldsymbol{w}) = \underbrace{\boldsymbol{Q}_t C_{t-1} \boldsymbol{Q}_t + \sigma^2 \boldsymbol{Q}_t \boldsymbol{S}_t^+ \boldsymbol{Q}_t}_{\boldsymbol{C}_t}.$$

This concludes the proof of Theorem 1. □

E.2 Proof of Theorem 2

Under the assumptions of the theorem, we have that

$$\begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix} \sim \mathsf{N}\left(\begin{bmatrix} \tilde{\boldsymbol{S}} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_* S_*^+ \end{bmatrix} (\boldsymbol{1}_{K+1} \otimes \theta), \sigma^2 \begin{bmatrix} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_*^+ \end{bmatrix} \right). \tag{5}$$

The goal for this proof is to verify that if $\boldsymbol{G}_1, \ldots, \boldsymbol{G}_t$ are full rank then

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_t] = \big(\mathrm{I} - \boldsymbol{Q}_t \cdots \boldsymbol{Q}_1 (\mathrm{I} - \boldsymbol{G}_0 \boldsymbol{G}_0^+)\big)(\boldsymbol{1}_K \otimes \theta), \quad \mathsf{Cov}(\hat{\boldsymbol{\theta}}_t) = \boldsymbol{M}_t \begin{bmatrix} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_*^+ \end{bmatrix} \boldsymbol{M}_t^\top + \boldsymbol{C}_t.$$

We proceed by mathematical induction. Consider the case $t = 0$. By (4) along with $\boldsymbol{Q}_0 = 0$, the mean is

$$\mathbb{E}[\boldsymbol{\theta}_0] = \boldsymbol{P}_t \begin{bmatrix} \tilde{\boldsymbol{S}} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_* S_*^+ \end{bmatrix} (\boldsymbol{1}_{K+1} \otimes \theta) = \boldsymbol{G}_0 \boldsymbol{G}_0^+ (\boldsymbol{1}_K \otimes \theta).$$

Likewise, recalling that $\boldsymbol{M}_0 = \boldsymbol{P}_0$, the variance is

$$\mathsf{Cov}(\boldsymbol{\theta}_0) = \boldsymbol{M}_0 \, \mathsf{Cov}\left(\begin{bmatrix} \tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}} \\ X_*^+ y_* \end{bmatrix} \right) \boldsymbol{M}_0^\top = \sigma^2 \boldsymbol{M}_0 \begin{bmatrix} \tilde{\boldsymbol{S}}^+ & 0 \\ 0 & S_*^+ \end{bmatrix} \boldsymbol{M}_0^\top.$$

Next, suppose that $\boldsymbol{G}_1, \ldots, \boldsymbol{G}_t$ are full rank and the stated distribution holds up to time $t-1$. By Theorem 1 we know that $\hat{\boldsymbol{\theta}}_t$ is Gaussian and so all that remains is to verify the given expressions for the mean and covariance. By the linearity of expectation and (4), the mean satisfies

$$\begin{aligned} \mathbb{E}[\hat{\boldsymbol{\theta}}_t] &= \boldsymbol{P}_t \begin{bmatrix} \mathbb{E}[\tilde{\boldsymbol{X}}^+ \tilde{\boldsymbol{y}}] \\ \mathbb{E}[X_*^+ y_*] \end{bmatrix} + \boldsymbol{Q}_t \mathbb{E}[\hat{\boldsymbol{\theta}}_{t-1}] \\ &= \boldsymbol{P}_t (\mathrm{I}_{K+1} \otimes \theta) + \boldsymbol{Q}_t (\mathrm{I}_K \otimes \theta) - \boldsymbol{Q}_t \boldsymbol{Q}_{t-1} \cdots \boldsymbol{Q}_1 (\mathrm{I} - \boldsymbol{G}_0 \boldsymbol{G}_0^+)(\boldsymbol{1}_K \otimes \theta), \end{aligned}$$

where the last follows from the inductive assumption applied to $\mathbb{E}[\hat{\boldsymbol{\theta}}_{t-1}]$. Moreover, from the definitions of $\boldsymbol{P}_t$ and $\boldsymbol{Q}_t$, we have

$$\begin{aligned} \boldsymbol{P}_t (\mathrm{I}_{K+1} \otimes \theta) + \boldsymbol{Q}_t (\mathrm{I}_K \otimes \theta) &= \boldsymbol{G}_t^+ \Big(\bar{\alpha}_t \beta_t \tilde{\boldsymbol{S}} + \bar{\alpha}_t \bar{\beta}_t (\mathrm{I}_K \otimes S_*)) + \alpha_t \Pi \boldsymbol{S}_t \Big) (\mathrm{I}_K \otimes \theta) \\ &= \boldsymbol{G}_t^+ \boldsymbol{G}_t (\mathrm{I}_K \otimes \theta) \\ &= \mathrm{I}_K \otimes \theta, \end{aligned}$$

where the second line follows from the identity $\Pi \boldsymbol{S}_t (\boldsymbol{1}_K \otimes \mathrm{I}_d) = (\mathrm{I}_K \otimes \underline{\boldsymbol{S}}_t)(\boldsymbol{1}_K \otimes \mathrm{I}_d)$ and the last line holds because $\boldsymbol{G}_t$ is full rank. Combining the above displays gives the desired expression for the mean. The expression for the variance follows directly from (5) and Theorem 1. □